

DNA of learning behaviors: A novel approach of learning performance prediction by NLP

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ABSTRACT

In recent years, the field of learning analytics has gained significant attention as educators and researchers seek to understand and optimize the learning process in online learning systems. This paper presents a novel methodology for predicting learning performance in online learning systems by leveraging natural language processing (NLP) and embedding techniques. The study focuses on two online learning systems, namely BookRoll and Viscode, and aims to analyze the learning behaviors of students using system logs extracted from the databases. The logs are converted into semester and daily learning description documents to capture the daily activities and progress of the learners. To transform the natural language data into numerical representations, the transformer-based BERT model, Google Gemini, and OpenAI large text embedding methodology are employed to generate embeddings for the learning descriptions. Subsequently, k-means clustering is applied to identify distinct learning behaviors exhibited by students. These clusters are labeled with numbers, and the daily learning descriptions are combined into a sequence, referred to as the DNA of learning behaviors. By utilizing this DNA representation, the learning status of students is effectively captured, and a machine learning model is trained to predict learning performance. The experimental results demonstrate the efficacy of the proposed methodology in achieving highly convincing predictions. The contributions of this research lie in the adoption of a unique approach, integrating NLP methodologies and embeddings techniques, to enable accurate learning performance prediction in online learning systems.

1. Introduction

In the seminal work of D.Bransford et al. (2000), they provided a comprehensive discussion of the learning research changed from speculation to science, and defined that learning involves the active construction of knowledge, skills, and understandings. It is not simply a process of passive absorption of information, but rather an active and dynamic process in which learners engage with content, make connections between new information and existing knowledge, and build conceptual frameworks that facilitate further learning and understanding.

In the dynamic landscape of education, where technology and data intersect, learning analytics emerges as a powerful lens through which we can interpret the complicated student interactions, optimize learning environments, and enhance educational outcomes. The roots of learning analytics trace back to early work on educational data mining (Dutt et al., 2017; Xu et al., 2021). However, it wasn't until the last decade

that learning analytics gained prominence. In 2011, the 1st International Conference on Learning Analytics and Knowledge defined learning analytics as the “measurement, collection, analysis, and reporting of data about learners and their contexts” (Siemens et al., 2011). Learning analytics aims to understand and optimize learning by analyzing data trails left by students and teachers. From understanding learning behaviors to predicting outcomes, learning analytics has become a pivotal field in educational research.

McNamara et al. (2017) discussed the framework and techniques of learning analytics which include the concepts and implementation approaches. Enabled by technological innovations, the teaching and learning process has undergone revolutionary changes. Massive open online courses (MOOCs) exemplify this shift, generating vast amounts of learning-related data. Learning analytics leverages big data to analyze student engagement, progress, and patterns. Educators now have near real-time insights into learning behaviors, enabling informed decisions and personalized interventions. Leitner et al. (2017) proposed the

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high-level process of learning analytics into different five phases of capturing, reporting, predicting, acting and refine. In our study, we focus specifically on learning behaviors analytics (LBA) within the broader scope of learning analytics (LA) where we addressed the key behaviors which composed by key activities and contribute to different learning performance result.

Learning behaviors refer to the actions, attitudes, and strategies exhibited by individuals during the process of acquiring knowledge, skills, or understanding. These behaviors can vary from person to person and can be influenced by factors such as motivation, environment, and individual learning styles. Learning behaviors encompass a wide range of activities, including paying attention, active listening, asking questions, taking notes, engaging in critical thinking, practicing, seeking feedback, and reflecting on the learning process. These behaviors are essential for effective learning and can enhance one's ability to acquire and retain new information. According to [Powell and Tod \(2004\)](#), they identified a set of learning behaviors drawn from the Qualified Teacher Status (QTS) professional standards (DfES/TTA 2002) in place at the time which includes engagement, interdependent activity, collaboration, responsiveness, participation, self-regard, communication, self-esteem, motivation and responsibility. In the book of "Learning and Behavior" by [Mazur \(2015\)](#), the author also discusses various learning behaviors that are observed in individuals. Some of the behaviors addressed includes classical conditioning, operant conditioning, observational learning, habituation and sensitization, and cognitive learning. Those studies discussed the different behaviors observed during the learning which provides the guidance of later learning behaviors research.

In the academic context, learning performance refers to the demonstration and application of acquired knowledge, skills, and understanding. It encompasses the extent to which an individual can effectively utilize what they have learned in various tasks, assessments, and real-world situations. Learning performance is typically assessed through various measures, such as exams, quizzes, projects, presentations, and practical demonstrations. It involves the application of learned concepts and skills to solve problems, analyze information, think critically, and communicate effectively. The academic performance of a learner is influenced by a range of factors, and one of the major factors is learning behaviors and as a result in bringing up attention of researchers to conduct performance prediction based on different learners' learning behaviors.

Predictive analytics with learning analytics involve using data, statistical algorithms, and machine learning techniques to forecast future outcomes based on historical data. The goal is to move beyond understanding what has already happened and provide a best assessment of what will happen in the future based on the learning behaviors analysis. The main objective of predictive learning analytics is to precisely predict the student's learning outcome regardless of all the challenges and limitation. The following activities after learning analytics always focus on tailoring learning experiences based on individual behaviors for personalization ([Dipace et al., 2018](#); [Li & Wong, 2020](#); [Wong et al., 2023](#)), identifying struggling students and providing timely support for early interventions ([Nimy et al., 2023](#)). The prediction results can also help shaping effective curricula through data-driven insights to achieve curriculum design or adaptive learning objectives ([Dennehy et al., 2020](#)). And based on the learning analytics process cycle ([Leitner et al., 2017](#)), the learning analytics will also focus on continuous improvement which enhancing teaching and learning effectiveness.

Predictive analytics encompasses various statistical techniques, including machine learning, predictive modelling, and data mining. The commonly used predictive models include decision tree, logistic regression, random forests, support vector machine (SVM), and neural networks ([Kumar et al., 2018](#)). The main task is to either perform classification or regression task to predict the student's learning outcome.

Despite the growing interest, there are still lots of challenges ahead. To ensure accurate and reliable data collection, Data quality remains a

hurdle from data quality perspectives ([Scheffel et al., 2014](#)). Besides, balancing data use with privacy and consent is always controversial from ethical concerns ([Tzimas & Demetriadis, 2021](#); [Weng, 2021](#)). In a typical research scenario of learning analytics, there is always gaps between data science, education, and psychology from interdisciplinary collaboration perspectives ([Blackmon & Moore, 2020](#), pp. 39–52). Ultimately, enhancing models to predict student outcomes accurately and effectively and seamlessly implement and integrate into pedagogical scenarios remains challenges. As we navigate these challenges, learning behavior analysis becomes a compass, guiding educators toward evidence-based practices and student success.

In this study, we adopt a unique approach to learning behavior analysis by integrating Natural Language Processing (NLP) techniques with embedding methodology and clustering algorithms. By leveraging NLP, we aim to extract rich insights from textual data, thereby capturing nuanced aspects of their engagement and interaction within educational contexts. The utilization of embedding methodology allows us to represent textual data in a high-dimensional space, facilitating the identification of semantic similarities and relationships among different learning behaviors. Moreover, by applying clustering algorithms to the embedded data, we can unveil hidden structures and patterns within the learners' behaviors, enabling the delineation of distinct behavioral profiles or segments for learning behaviors DNA (Deoxyribonucleic acid) generation.

This approach not only offers a comprehensive understanding of learning behaviors but also provides actionable insights for personalized learning interventions and instructional design optimizations. By integrating these methodologies, we present a novel framework that contributes to advancing the field of learning behavior analysis and learning performance prediction, offering new perspectives and opportunities for research and practice in educational settings. Based on this, the following three research questions we will answer.

RQ1: Can natural language description of learning behaviors contribute better predictive analytics result?

RQ2: Can learning behaviors be categorized by natural language description?

RQ3: How learning behaviors DNA will impact the learning outcomes?

In the remaining sections of this paper, we will provide literature review with related predictive learning analytics research in Section 2. The methodology used in this study will be addressed in Section 3, include datasets, data pre-processing steps, embedding and clustering methodologies, and machine learning models. The result and discussion will be in Section 4 where we will discuss the findings and observations of the outcome to answer all the research questions above. The conclusion and potential future research areas will be shared in Section 5.

2. Literature review

In the realm of education and cognitive science, understanding the intricate dynamics of learning behaviors has emerged as a paramount area of investigation. Learning behaviors encompass a broad spectrum of cognitive, affective, and metacognitive processes that individuals employ during the acquisition of knowledge. This domain has brought up significant attention from researchers seeking to unravel the complexities of human learning, with a particular focus on identifying patterns, strategies, and influential factors that shape learning outcomes. [Kuo et al. \(2021\)](#) designed a Moodle plug-in that can visualize students' learning behavior patterns in a Moodle course and cluster them into different groups based on their patterns. The plug-in allows teachers to see the relations among the learning objects in a course, review the individual and common learning behaviors of students or groups, and annotate the learning patterns based on their observations.

There are typical approaches of learning behavior analysis. Descriptive analysis involves summarizing and describing learners' behavior patterns such as frequencies, averages, and distributions, are often used to characterize various aspects of learning behavior, such as time spent on tasks, interaction patterns, navigation paths, etc. (Fisher et al., 2021). Sequence analysis focuses on analyzing the sequential patterns of learner interactions (Yang & Ogata, 2023). This approach is particularly useful for understanding the sequence of actions taken by learners as they progress through learning tasks or modules. Techniques such as sequential pattern mining, Markov models, and hidden Markov models can be applied to identify common sequences of actions and transitions between states.

The study of Geng et al. (2024) used lag sequential analysis (LSA) to examine the learning behavioral patterns and self-regulated learning (SRL) strategies of high school students who used an e-book system for a mathematics course. The study found that students used rehearsal and elaboration strategies, but lacked metacognitive strategies, in their e-book learning process. The study also revealed that students' performance varies from different learning behaviors during learning. Yang and Ogata (2023) reveals 15 sequential behaviors of e-book note-taking that indicate the cognitive strategy use of SRL, such as adding or changing memos after turning pages, deleting markers after adding memos, and adding bookmarks after adding markers. The study also shows the relationship between learning behaviors and learners' final achievement. These studies offer opportunities for educators to understand learners' learning behaviors and provide evidence-based feedback and interventions to improve course achievement in learning scenario.

Clustering analysis is used to group learners based on similarities in their behavior patterns. This approach helps identify distinct learner profiles or segments with similar characteristics or engagement patterns. Clustering algorithms such as K-means clustering, and hierarchical clustering can be applied to partition learners into meaningful groups (Mihăescu, 2011; Yathongchai et al., 2013).

There are some other approaches like Social Network Analysis (SNA) examines the social interactions and relationships among learners within educational settings where Cela et al. (2015) conducted a systematic review and discuss the related studies. Clickstream analysis focuses on analyzing the sequence of clicks and interactions made by learners within digital learning environments which help the researchers identifying the learners activity patterns and potentially can contribute to personalized learning and intervention (Filvà et al., 2019; Goulden et al., 2019). Sentiment analysis involves analyzing text data to understand learners' attitudes, emotions, and opinions. Natural language processing (NLP) techniques are applied to extract sentiment polarity, topics, and themes from textual data (Dharme & Karmore, 2022). Time series analysis is used to analyze temporal patterns and trends in learners' behavior over time. This approach allows researchers to identify recurring patterns, seasonality, and long-term trends in learning behavior metrics such as activity levels, engagement rates, and performance indicators. These approaches can be used individually or in combination to analyze various aspects of learning behavior and gain insights into learners' interactions, engagement levels, and performance within educational contexts.

In the ever-evolving landscape of education and technology, predictive learning analytics emerges as a powerful tool to enhance student outcomes, optimize educational processes, and inform decision-making. At its core, predictive learning analytics leverages data, statistical algorithms, and machine learning techniques to identify the likelihood of future outcomes based on historical data. Sghir et al. (2023) select and analyze 74 relevant publications from 2012 to 2022 and the findings show that predictive learning analytics (PLA) is mainly used to predict student performance and at-risk students, and that artificial neural networks, random forest, and gradient boosting are the most common machine learning methods. The findings also reveal the benefits and limitations of PLA in terms of data availability, quality, privacy, and explainability.

Predictive analytics encompasses a variety of statistical techniques, including machine learning, predictive modeling, and data mining. It draws from both historical and current data to estimate future behaviors, such as student performance, retention rates, and course completion. Namoun and Alshantiti (2020) conducted a systematic review to analyze 62 relevant studies published between 2010 and 2020 and found that the learning outcomes were predicted mainly as performance classes and achievement scores, using regression and supervised machine learning models. Student online learning activities, term assessment grades, and student academic emotions were the most evident predictors of learning outcomes. While predictive models play a central role, machine learning algorithms drive the process, enabling computers to learn from data without explicit programming. As these algorithms become more intelligent, they empower decision-makers with accurate and data-driven insights (Sghir et al., 2023).

While machine learning is a dominant approach in predictive learning analytics, there are other methodologies that don't rely on machine learning techniques. Rule-based models use predefined rules or decision trees to predict outcomes. These rules are often based on expert knowledge or domain-specific guidelines. For instance, in the context of student retention, a rule-based model might predict that if a student attends more than 80% of classes, submits assignments on time, and actively participates in discussions, they are likely to succeed (Sghir et al., 2023). The clustering technique is commonly used to group similar students based on shared characteristics to identify student segments and association rule mining identifies patterns in student behavior which may reveal interesting relationships, such as students who frequently visit the library tend to perform better. Chembu and Sanner (2023) proposes a generalized optimization framework for predictive clustering that combines different cluster definitions and supervised objectives. The other techniques like Principal Component Analysis (PCA) (Shlens, 2014) reduce the dimensionality of data while preserving important information. By reducing features, educators can focus on key variables that impact student success.

Remember that these non-machine learning approaches have their own advantages and limitations. While machine learning models often outperform rule-based or non-parametric methods, the latter can be more interpretable and easier to implement in certain educational settings. Machine learning plays a pivotal role in predicting academic outcomes, enhancing educational processes, and empowering decision-makers. In the review of (Sghir et al., 2023), artificial neural networks, random forest, and gradient boosting are the most common machine learning methods. (McCarthy et al., 2019) claimed that decision trees are simple yet powerful models that segment data into branch-like segments based on input variables. They help understand decision paths and can predict student behaviors, such as whether a student will drop out, succeed, or need additional support. Random forests are ensembles of decision trees. They combine multiple trees to improve accuracy and robustness. Random forests can predict student performance, recommend personalized learning paths, and identify critical features (Jin, 2023; Svetnik et al., 2003).

Another technique to enhance Decision Trees is Gradient Boosting. Gradient boosting builds an ensemble of weak learners (usually decision trees) sequentially, correcting errors made by previous models. Natekin and Knoll (2013) discussed how gradient boosting uses an iterative procedure that combines weak models, called base-learners, into a strong ensemble model. It excels in predicting student success, identifying at-risk students, and optimizing resource allocation. Some other common machine learning techniques like Support Vector Machines (SVM) (Chen & Zhai, 2023; Cortes & Vapnik, 1995) and Logistics Regression (Chen & Zhai, 2023; Sghir et al., 2023) can also predict the student performance outcomes based on input features.

Over the last decade, the growth in Artificial Intelligence (AI) has accelerated the capacity to capture and gain insights from various aspects of the learning experience. Predictive learning analytics dramatically adopted Deep Learning techniques to learn complex patterns from

data. They consist of interconnected layers of artificial neurons. Deep learning models can predict student performance, recommend personalized content, and analyze unstructured data (e.g., essays) (Sghir et al., 2023).

Predictive learning analytics also leverages NLP techniques to extract meaningful information from text data. Researchers have identified valuable text data assets within educational contexts (Feder et al., 2022; Khurana et al., 2023; McNamara et al., 2017). These include student essays, discussion forum posts, course materials, and administrative documents. This kind of information is treated as text data assets which is the key foundation of predictive learning analytics in NLP. Zhang et al. (2018) leverage NLP as a tool for sentiment analysis to gauge student emotions, engagement, and satisfaction, and other functions like topic modeling to uncover latent themes in student discussions, identifying common topics and trends (Feder et al., 2022; Khurana et al., 2023), and text classification to classify student essays or responses into predefined categories (e.g., high-quality vs. low-quality) (Lodhi et al., 2002).

There are lots of benefits by using NLP into predictive learning analytics, Allen et al. (2022) discussed the general guidance of using NLP in learning analytics and can be combined with other data sources, such as eye-tracking, keystroke logging, and affect detection, to provide a more comprehensive and dynamic view of the writing process and its outcomes. The NLP-powered models can be an early warning system (Akcapinar et al., 2019; Suahati et al., 2023) to predict at-risk students, allowing timely interventions. For instance, identifying students expressing negative sentiments or discussing academic struggles can trigger personalized support. Besides, the NLP can also assist in providing personalized feedback (Kochmar et al., 2020) to students like automated feedback on writing assignments, discussion posts, or quizzes enhances the learning experience. Additionally, NLP-driven content recommendation (Khurana et al., 2023) systems can also suggest relevant resources based on student queries or interests.

In addition to the challenges mentioned in the Introduction section, using NLP will bring more challenges. NLP models heavily rely on data quality. Noise, misspellings, and ambiguity in student-generated text can affect accuracy (Shaik et al., 2022; Susnjak, 2023). One of the other challenges is handling diverse languages and dialects poses challenges (Abe et al., 2020; Duong, 2017) and thus NLP models must adapt to different linguistic minor differences. Finally, NLP models should respect privacy, avoid bias, and maintain fairness which is also a common challenge area when bringing AI into the system implementation.

3. Methodology

In this section, we outline the systematic approach shown in Fig. 1 to extract meaningful insights from our data. This section encompasses three critical stages: data sources, data pre-processing, and training two models focusing on the entire semester and daily learning descriptions. The data sources contain instructional online systems, while we designed a natural language converter to convert systems logs into natural languages. Model 1 and Model 2 are two separated machine learning models for learning performance prediction. As for the learning

behaviors clustering and learning behaviors DNA, we leverage different embedding techniques and clustering algorithms. We will discuss the details in the following sub-sections.

3.1. Data source

Our primary data source comprises student enrollment records from the Python programming course of the university spanning the years 2020–2022. Within this timeframe, we collected information on 465 students from 5 semesters and 9 classes without prior computer science domain knowledge. All the students were aware of the learning analytics research and have the consent of collecting raw data from the system for further analysis. By focusing on this specific backgrounds outside of IT, we aim to uncover learning patterns and performance prediction. The students' data includes attributes such as student IDs, enrollment dates, and academic achievements.

Our secondary data source stems from two online platforms: BookRoll and Viscode (Huang et al., 2020). The decision to utilize data from both systems stems from the recognition of the multifaceted nature of learning behaviors in programming language courses. BookRoll offers a rich repository of user interactions with electronic textbooks, encompassing engagement metrics such as reading duration, bookmarking frequency, and annotation activity. By leveraging BookRoll data, we aim to gain insights into learners' study habits, content preferences, and comprehension strategies within the context of programming course materials. Complementarily, data from Viscode provides a window into learners' practical application of programming concepts and problem-solving approaches. Through the analysis of coding sessions, error logs, and debugging processes within the Viscode environment, we seek to uncover patterns of learning progression, coding proficiency development, and misconceptions resolution. By integrating data from both BookRoll and Viscode, this study endeavors to offer a comprehensive understanding of the interplay between theoretical knowledge acquisition and practical skill development in programming education. The synergy between these two data sources facilitates a holistic exploration of learning behaviors, thereby enriching our insights into effective pedagogical strategies and personalized learning interventions for programming language learners. The online system log data will then be merged and translated to natural language documents grouped by semesters and days which will be explained in Section 3.2. The total records are shown in Table 1.

Table 1
Data source years distribution and records count.

	Year 2020		Year 2021		Year 2022
	Semester1	Semester2	Semester1	Semester2	Semester1
Month	SEP-JAN	MAR-JUN	SEP-JAN	MAR-JUN	SEP-JAN
Student	53	119	140	41	112
BookRoll	247,628	568,193	640,895	196,797	268,681
Viscode	142,638	436,145	441,917	277,486	567,106
Total Logs	390,266	1,004,338	1,082,812	474,283	835,787

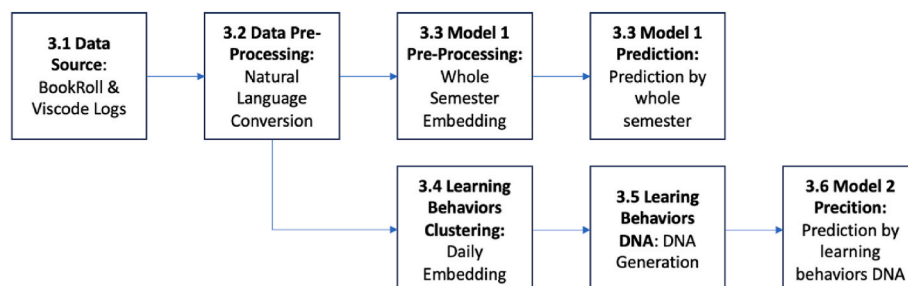


Fig. 1. Research methodology process.

The rich dataset will serve as a foundation for understanding the student behavior, learning performance prediction, and potential interventions.

3.2. Data source pre-processing

The purpose of data pre-processing is to convert BookRoll and Viscode system logs into natural language sentences. We began by extracting all system logs into Python Pandas DataFrame where each row corresponds to a single log entry. The DataFrame structure allows us to manipulate and analyze data efficiently where we merged those redundant records before converting to sentences. To make the logs more human-readable, we establish conversion rules. These rules map specific log events to natural language sentences. For instance, if a log entry indicates that a student accessed a particular page and made a highlight, we convert it to a sentence like: “The student read from page 1 to 5 of C1 for 70 s and added marker at page 5”. The other example is if a log entry indicates that a student try to write some code in Viscode and get the error, we convert it to a sentence like: “The student is coding and execute the codes with error SyntaxError for 200 s, and with code length 49”. Similarly, we handle other log types (e.g., create memo, turning page, coding without error,...etc) and create descriptive sentences for each. By doing so, we bridge the gap between raw data and meaningful insights, and further can be processed by text embedding models.

The raw data also include scores of performance metrics. To simplify interpretation, We categorize these scores into three level.

- **High:** Represented by the top 25% of scores.
- **Medium:** Encompassing the middle 50% of scores.
- **Low:** Comprising the bottom 25% of scores.

The transformation allows us to analyze learning outcomes more intuitively. All the sentence-based information is segregated by either the whole semester or by day, which will be used by different model and we will explain that in later sections. The total number per semester is not identical to the 465-student count because some of the students did not interact at all with systems during the semester. The converted logs into natural language sentences is shown in Table 2.

In summary, the data pre-processing pipeline ensures that we convert raw logs into coherent natural language sentences, merge redundant entries, and transform scores into three categories. These processed logs will serve as the foundation of training data set for subsequent model training.

3.3. Model 1: learning performance prediction by semester

We explored 3 popular text embedding methods to convert semester based learning description into embeddings, they are from HuggingFace pretrained BERT (HuggingFace, 2024), Goggle Gemini AI (Google, 2024), and OpenAI embedding service (OpenAI, 2024). These 3 embedding methods have different variants to meet various requirements. The summary information of different models is shown in Table 3.

The deep learning model within Model 1 and Model 2 defines a simple linear model as a class implemented using PyTorch, a popular deep learning framework. The purpose of this model is to perform learning behaviors analysis, which typically involves predicting an output based on input features related to learning behaviors.

Table 2
Sentence segregation by semester and day.

	Total Sentences	Grouped by Semester	Grouped by Day
BookRoll	146,210	458	11,180
Viscode	243,671	449	9229
BookRoll + Viscode	389,881	458	14,066

Table 3
Text embedding models summary.

Supplier	Model Name	Input Token Limit	Output Dimensions
HuggingFace	bert-base-cased	512	768
Goggle Gemini	models/embedding-001	2048	768
OpenAI	text-embedding-3-large	8191	3072

The linear model class consists of three linear layers defined sequentially with rectified linear unit (ReLU) activation functions applied after the first two linear layers. The final output of the model is obtained from the last linear layer. In essence, this model aims to learn relationships between input features related to learning behaviors and predict an output relevant to the analysis, leveraging a simple linear structure with ReLU activations. The model framework is shown in Fig. 2.

As the first step, we use the 3 embedding models above to embed the texts and the embedded semester learning description documents are then passed to a feedforward linear model as training data for learning performance prediction with categorized performance score as labels. We adopted K-fold training methodology with $K = 5$ and randomly allocated 80% of data as training set, and the remaining 20% as validation set. The loss function in this model is MSE with learning rate = 0.001, and the optimizer we used is RMSprop. There are 500 epochs in each fold, and we will show and explain the result in later sections.

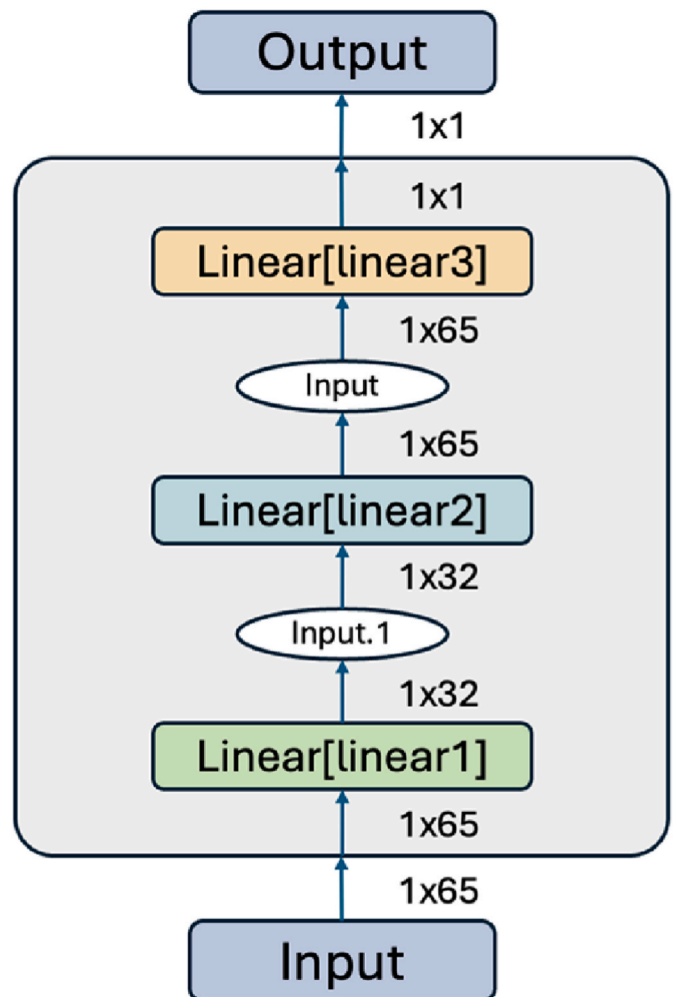


Fig. 2. Model 1 and model 2 deep learning framework.

3.4. Learning behaviors clustering

In this study, we delve into the intricate world of embedding techniques and clustering algorithms to uncover latent patterns within educational documents. Our goal is to understand the underlying structure of textual data related to learning description documents, specifically focusing on the BookRoll, Viscode, and BookRoll + Viscode documents.

To transform raw text into meaningful numerical representations, we employ state-of-the-art embedding models. Specifically, we utilize three prominent embeddings including HuggingFace pretrained BERT (Devlin et al., 2018; HuggingFace, 2024), Google Gemini API (Google, 2024) and OpenAI embedding service (OpenAI, 2024). The summary information of these embedding models can be found in Table 3. We apply these embeddings to our semester-based and daily-based documents, aiming to capture nuanced semantics and contextual cues.

Once we have embedded our documents, we proceed with K-Means clustering (MacQueen, 1967), a popular unsupervised learning technique. K-means partitions data into clusters based on similarity. Our choice of the number of clusters significantly impacts the quality of results. We evaluate different cluster counts using the silhouette score, which balances cohesion within clusters and separation between clusters. The silhouette score guides us in selecting the optimal number of clusters.

To gain insights into the clustered data, we employ T-SNE (t-Distributed Stochastic Neighbor Embedding). T-SNE reduces high-dimensional embeddings to a lower-dimensional space while preserving pairwise similarities. By visualizing the clustered data in this reduced space, we aim to reveal meaningful groupings and potential outliers. In summary, our methodology combines cutting-edge embeddings, K-Means clustering, and T-SNE visualization to unravel hidden structures within educational documents.

3.5. Learning behaviors DNA

In our quest to decode the student interactions within educational ecosystems, we introduce the concept of Learning Behaviors DNA. This metaphorical construct allows us to distill raw behavioral data into meaningful patterns, akin to genetic sequences. We begin by converting daily learning behaviors into natural language sentences and then passed the sentences into pre-trained BERT to get embeddings. In Section 3.4, the embeddings can be clustered by using K-Means and mapped each embedding into relative labeled values.

Just as DNA strands string together nucleotides, we concatenate daily behavior labels into a sequence. If there is no any system activity for a specific day, we add an additional number value to ensure all the students' sequences has the same length. This sequence becomes the student's Learning Behaviors DNA—a compact representation of their semester-long engagement pattern. The DNA sequence may look like the following Fig. 3.

Our Learning Behaviors DNA can be treated as a vector. Each position in the vector corresponds to a day, with values representing the behavior clustering from the BERT embedding. This vector encapsulates

	username	learning_behavior_DNA	level
0	[REDACTED]	[0, 2, 0, 0, 0, 0, 1, 1, 1, 1, 2, 1, 0, 0, 3, ...	1
1	[REDACTED]	[2, 2, 1, 1, 0, 1, 1, 1, 3, 3, 3, 3, 3, 3, ...	2
2	[REDACTED]	[0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, ...	2
3	[REDACTED]	[2, 0, 0, 0, 0, 0, 0, 0, 1, 2, 0, 2, 0, 0, 0, ...	1
4	[REDACTED]	[2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...	2

Fig. 3. Learning behaviors DNA

the student's entire semester of learning interactions with BookRoll and Viscode. By embedding it into a continuous space, we capture the essence of their engagement trajectory. Just as genetic variations influence health outcomes, our Learning Behaviors DNA correlates with academic performance and will be used to train our Model 2.

3.6. Model 2: prediction by learning behaviors DNA

We train the predictive model by arming it with Learning Behaviors DNA vectors and performance labels. Our goal is to obtain a model which can predict the students' performance, just as genetic testing informs health risks, our model anticipates academic trajectories. The Model 1 and Model 2 are using the same feedforward model, and the only difference is the input dimensions.

The concept of creating a "learning behaviors DNA" is interesting and inspiring. Just as our genetic DNA encodes essential information about our physical traits and health, this learning behaviors DNA encapsulates critical aspects of student learning. Just as variations in human DNA impact health outcomes and behaviors, variations in learning behaviors DNA could significantly influence academic performance. Our methodology's strength lies in its ability to predict student performance based on this learning behaviors DNA. By quantifying and encoding learning description documents into a sequence of numbers, we are unlocking valuable insights.

In summary, Learning Behaviors DNA bridges the gap between raw logs and actionable insights. By understanding these patterns, educators can tailor interventions, personalize learning experiences, and cultivate student success. We will share the training result and discussion in Section 4.

4. Results

In this section, we present the outcomes of our comprehensive predictive analytics approach applied to learning analytics. Our methodology encompasses two distinct models: Model 1, which leverages the entire learning description document embedded by different models for predicting learning performance, and Model 2, which incorporate learning behaviors DNA as the daily learning activities representation for prediction. We will also discuss how learning behaviors DNA will impact the learning outcome in the next section.

4.1. Model 1 performance prediction

The Model 1 leverages the embedding from 3 different document embeddings models as the features in the training document. The objective is to predict the learning outcomes of High, Medium, and Low. As mentioned in the methodology section, we adopted K-fold approach to separate training and validation set. The overall training involves BookRoll, Viscode and the consolidation of BookRoll and Viscode. Below Table 4 is the training result scores.

The result of OpenAI embedding is quite encouraging and convincing and achieves very high accuracy of learning outcome prediction. Below

Table 4
Training result with validation set scores.

Embedding	Data Source	F1	Recall	Precision	Accuracy
BERT	BR	0.21	0.33	0.46	0.46
	VS	0.25	0.33	0.61	0.61
	BR + VS	0.31	0.37	0.47	0.48
Gemini	BR	0.91	0.91	0.90	0.90
	VS	0.72	0.71	0.75	0.75
	BR + VS	0.81	0.78	0.86	0.80
OpenAI	BR	0.95	0.93	0.97	0.96
	VS	0.98	0.99	0.97	0.98
	BR + VS	0.98	0.99	0.97	0.98

BR: BookRoll, VS: Viscode, BR + VS: Consolidation of BookRoll and Viscode.

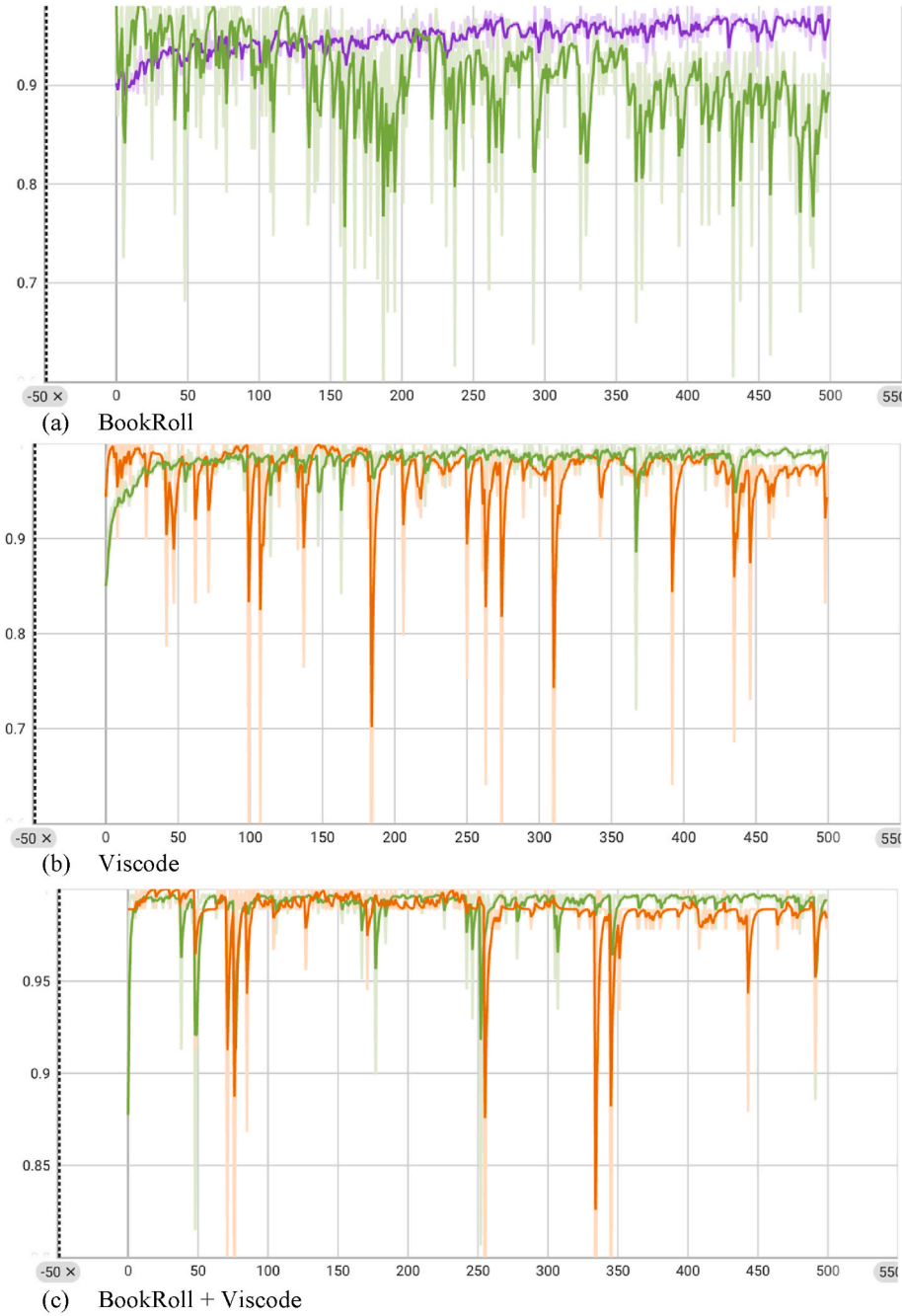


Fig. 4. OpenAI embedding training result.

Fig. 4 is the training and validation chart of OpenAI embeddings during the training for BookRoll, Viscode and both. Here we only show the last 500 epochs of the K-fold training.

One of the major reasons why OpenAI is outperforming among these 3 embedding methods is because OpenAI embedding input tokens can be up to 8191 while Google Gemini embedding can only accept 2048 token and BERT can only accept 512 tokens. The intuitive observation is that the more content the embedding method can include, the more meaningful semantics can be included in the output embedding vectors.

4.2. Learning behaviors clusters

To observe if there is any pattern between the learning description and performance result, we utilize the models in Table 3 to get semester and daily embeddings. We applied the embeddings from 3 different

models into sklearn kmeans clustering and use silhouette score to verify clustering result. However, the semester-based embeddings clustering silhouette scores were all < 0.4 which may not be suitable to perform the next steps of **Model 2** training.

We then further checked the daily learning descriptions embeddings from the models again and found the clustering results from Hugging-Face pre-trained BERT is quite good. The other 2 models clustering result silhouette scores are again always less than 0.4 and may not be suitable again to proceed the next steps. The reason might be because the average tokens of daily learning description document is normally less than 512 tokens where BERT can perfectly cover the whole text embedding. While Google Gemini and OpenAI embedding service need to pad lots of empty tokens to make token length which results in most of the text content are padding tokens and cannot be differentiated after converted to embedding vectors.

The different embedding methods with different data source matrix can be shown in the following Fig. 5. We used T-SNE methodology to visualize the embedding result. Besides, when we looked at the best silhouette score from each embedding methods, BERT embedding result was much better than the other two methods. Based on the clustering result, we used HuggingFace pre-trained BERT embedding to categorize different learning behaviors on the daily basis. The BookRoll daily learning activities will be categorized into 3 types, labeled as 0 to 2. The Viscode daily behaviors can also be categorized into 3 types and same as the consolidation of BookRoll and Viscode. The behaviors will be labeled as 0, 1, and 2 with different behaviors meaning which will be explained in the following paragraphs. We added another label 3 which identifies when there is no any activities and interactions with the system and simply refers to the student did nothing during the day. The purpose of doing this is to ensure later on the total length of learning behaviors DNA sequences among the students will be consistent.

4.3. Model 2 performance prediction with learning behaviors DNA

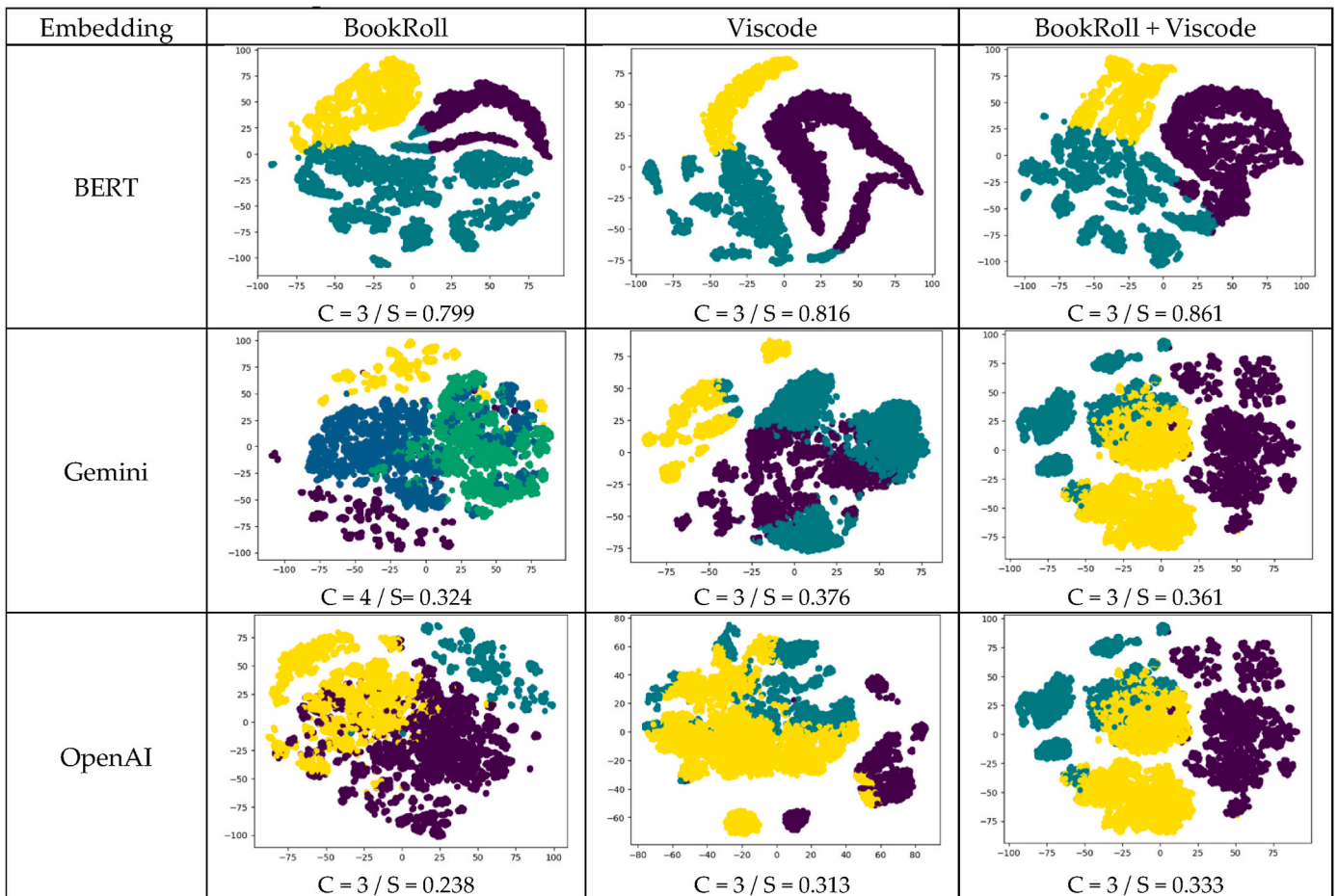
Model 2 takes the learning behaviors DNA as the features input which represents the daily learning behaviors of the whole semester. Since only BERT embedding result can be qualified by silhouette scores, we use the cluster result labels to concatenate as the learning behaviors DNA for Model 2 training. The labels of the training data are the same with Model 1 which is the categorized semester scores which is 0, 1, or 2. The training result with scores can be seen in the following Table 8. When we use BookRoll daily learning activities, the result is not

convincing (F1 score is 0.55). However, when including Viscode and combine BookRoll and Viscode together, the training result is convincing. This may because of daily BookRoll activities can not differentiate the students' behaviors while Viscode daily activities provide more diversity of different activities.

We use K-fold method like Model 1 training with the same hyper parameters and feedforward model, the chart of training and validation of 5th K-fold training process can be seen in the following Fig. 6. For BookRoll and Viscode consolidation training, the F1 score is 0.95 with recall 0.95 and precision 0.95 respectively. The F1 score of purely using BookRoll daily learning behaviors seems not high but Viscode learning activities contribute a lot when consolidating both daily activities together.

4.4. Learning behaviors DNA

A noteworthy observation emerges when we average the learning behaviors DNA of students categorized as high, medium, and low performers, and the patterns is illustrated in Fig. 7. The pattern shows how different the learning behaviors during the classes and out of classes. High-scoring students tend to exhibit consistently active learning behaviors, typically with a learning type index of <1 (indicative of active learning). In contrast, low-scoring students show an average learning type index of >1.5 . This suggests that high-scoring students align their learning pace with the instructor's, engaging actively with Bookroll ebooks and practicing coding in Viscode accordingly. When examining behaviors outside of class time, we found the behaviors are not quite



Note: C: Clusters, S: silhouette score

Fig. 5. Bert clustering results with T-SNE visualization.

Note: C: Clusters, S: silhouette score.

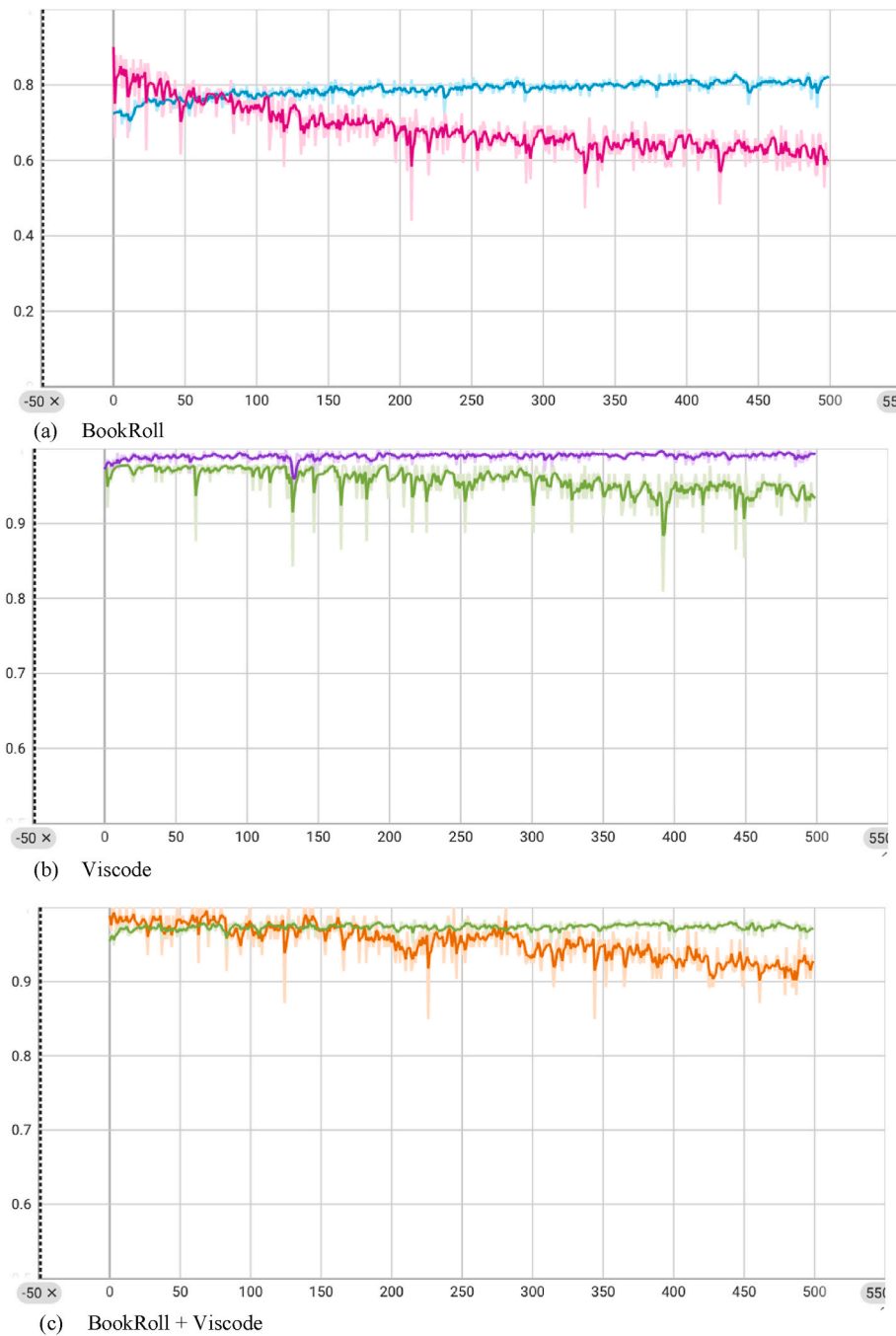


Fig. 6. Training result with learning behaviors DNA

consistent between different groups of students. However, a notable difference arises toward the end of the final examination period of the semester, high score students learning behaviors is more active than medium and low score students. In summary, students who are actively engaged during class and dedicate more effort before examinations tend to achieve better learning performance outcomes.

From the patterns observed, it can be deduced that students tend to adhere closely to the instructional cycle provided by the teacher, indicating a significant alignment with the principles outlined in the forgetting curve theory (Ebbinghaus, 2013). This theory posits that the retention of learned material diminishes over time without active reinforcement or review. In this context, students' heightened engagement during class time, particularly when following the instructor's progress and interacting with resources such as BookRoll and Viscode,

corresponds to the initial stages of learning when information is most readily absorbed. Moreover, the finding that high-scoring students exhibit greater self-discipline by engaging in more review activities aligns with the notion that regular review counteracts the effects of the forgetting curve, thereby facilitating long-term retention and mastery of the material. This convergence between observed learning behaviors and the principle of the forgetting curve underscores the importance of strategic intervention strategies aimed at optimizing learning outcomes and mitigating the natural decay of learned information over time.

5. Discussion

In this section, we discuss the outcome of two models' performance prediction, learning behaviors clustering, and the implication of

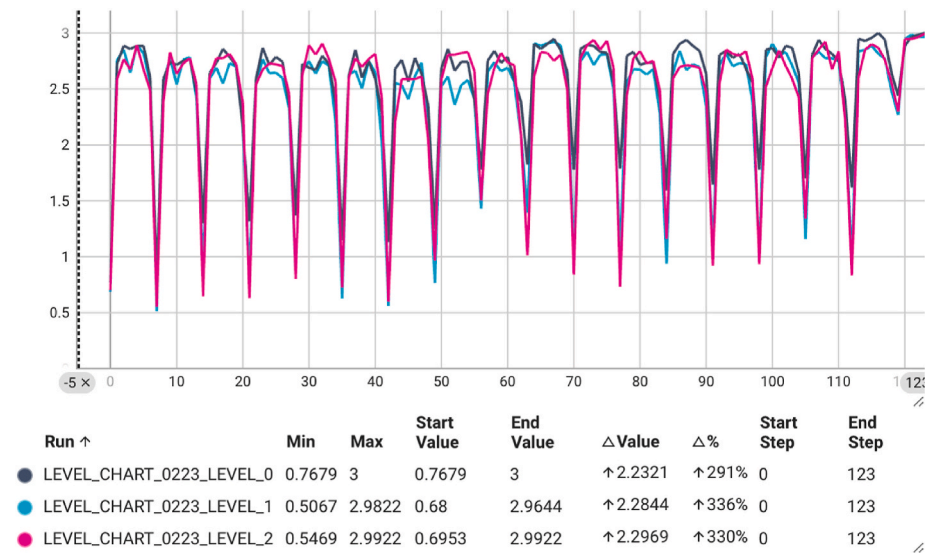


Fig. 7. Daily average of learning behavior types.

learning behaviors DNA. We also response to the research questions one by one which listed in the introduction section.

5.1. Model 1 prediction and response to RQ1

In the review of (Albreiki et al., 2021), the authors present the results and discussion of the literature analysis in three sections: predicting the performance of students at risk, students dropout prediction, and evaluation of students' performance based on static and dynamic data. The review compares the approaches, datasets, features, and algorithms used in different studies, and highlights the gaps and challenges in the current research. The review follows Okoli's guide for conducting a standalone systematic literature review, which involves identification, screening, eligibility checking, and inclusion criteria meeting of the research papers. The review also defines four quality criteria for the selected papers: clear review objectives, well-defined methods, validated accuracy, and explicit limitations.

Compare with the research of (Li et al., 2024), we employ the similar data sources but using different prediction methodology. The previous study focused on the prediction of logistic regression and neural network with explainability while our research focus on final score prediction by using NLP. The result from (Li et al., 2024) shows that explainability of LIME is better for logistic regression and SHAP is better for neural network. Although the previous research offers valuable insights into student learning behaviors and strategies, it is essential to consider the prediction accuracy achieved in both the study and our current work. Imran et al. (2019) uses a dataset of 1044 students from two schools in Portugal, and applies three classification algorithms (J48, NNge and MLP) with a Realadaboost meta classifier. The results show that the J48 classifier with Realadaboost achieved the highest accuracy of F1-Score 95.78%. In the study of (Sekeroglu et al., 2019), the authors describe two groups of prediction experiments using the Student Performance Dataset (SPD), which contains student grades for Maths and Portuguese courses. The authors use three machine learning algorithms: Back-propagation (BP), Support Vector Regression (SVR), and Long-Short Term Memory (LSTM), and evaluate them based on three criteria: Mean Square Error (MSE), R2 Score, and Explained Variance (EV) Score. The authors find that SVR achieves the highest prediction rates and lowest error rates for both courses, followed by LSTM and BP.

Given the novel methodologies in our study, we achieved higher and better prediction accuracy. Besides, by thoroughly examining the prediction performance in each study, we can gain a deeper understanding of the effectiveness of our approaches. In our research, we have achieved

better accuracy in predicting performance outcomes. This comparison underscores the importance of carefully evaluating prediction performance in educational data mining studies to ensure the reliability and effectiveness of the proposed methodologies.

Thus, to reply to **RQ1**, the natural language processing can be one of the effective approaches to predict the students' learning outcomes. This approach does not only provide better learning performance prediction but also better interpretability since the behaviors are described as natural language. This approach can also improve features representative since some key features may be eliminated through features engineering while large text embedding will have a holistic context of the learners activates.

5.2. Learning behaviors clustering and response to RQ2

If we examine at the clustering results from BookRoll training set, the overall summary is shown in the following Table 5. The average length means the length of the daily learning description document, average module means how many modules of the learning materials were covered, and average time means how much time the learners spent on BookRoll system. Among these 3 categories, the category ID 0 describes that the learners' have many reading activities with highest modules (6.19) covered, while the average spent time is the lowest among these 3 categories. This means the learners browse the materials quickly without spending too much time in the detailed. The category ID 1 has the least activity description and modules coverage, but with the longest time spent. This indicates that this kind of learners maybe idle most of the time. The category ID 2 is the balance type between category ID 0 and 1, which means this kind of users have medium size of learning activates and spend reasonable time with module coverage. Thus, we identified this kind of learns as detailed learners.

Table 5
BookRoll learning behaviors clustering.

ID	Count#	Avg. Length	Avg. Module#	Avg. Time	Comments
0	2394	494.90	6.19	1106	Quick browsing learners
1	6452	48.04	2.29	3760	Less efficient and idle learners
2	3034	130.54	4.83	2353	Detailed learners
3	–	–	–	–	No activities during the day

Below in Table 6 is the clustering result of Viscode learning activities. The category 0 is the active learners with lots of activities since the length of learning description document is long while the average time spent is 1204 s. Following the same observation concept, we can find category 1 users are inactive without too much efficiency since there is not too much activities but the overall time spent is long with 9733 s. We also calculate the average program execution error rate in the learners' daily activates, and found there is no big difference between these three categories.

Similarly, if we look at the clustering result of the consolidation of BookRoll and Viscode, the clustering result is shown in Table 7. Between the average learning description document length and the time spent on the daily basis, we can conclude the 3 different users learning behaviors as high, medium and low efficiency learners. For example, category 0 users have long learning description document (average length is 671 words) but with less time spent (average 13,132 s).

Compare with other research, Yathongchai et al. (2013) collects data from Moodle learning management system, which records the students' activities and interactions in an e-learning course and the MIS system of the university, which records the students' grades and scores in four compulsory computer subjects. This study applies K-means algorithm to cluster the students into four groups based on their learning profiles, which consist of eight activities such as frequency of login, resource view, document download, etc. The result describes the characteristics of each cluster, such as the lazy students, the skill-practicing students, the reading students, and the assiduous students. Rao et al. (2016) uses a dataset of 200 records from rural-based computer science and engineering students, and performs data preprocessing by converting attributes to categorical, reducing dimensionality, and handling missing values. This study finds that Random Forest algorithm shows better accuracy than J48 and Naïve Bayes as the dataset size increases, and that providing good study material and lab work can improve the performance of poor students.

In summary, addressing RQ2, the findings highlight the efficacy of leveraging BERT embeddings for clustering daily learning behaviors. By harnessing the semantic richness encapsulated within BERT embeddings, our approach effectively categorizes learning behaviors into distinct clusters, each encapsulating unique patterns in learning behavior. Compare with traditional methods reliant on simplistic feature representations, BERT embeddings offer a nuanced understanding of the underlying semantic structures present within learning description documents. The utilization of K-means clustering further enhances this process by uncovering latent patterns in an unsupervised manner, thereby providing insight into the diverse spectrum of learning behaviors. This integration of advanced language representations and scalable clustering techniques not only ensures flexibility and scalability but also fosters interpretability, enabling stakeholders to discern meaningful insights from the resulting clusters. Moreover, the robustness of BERT embeddings, stemming from their pre-training on vast text corpora, underpins the reliability of our clustering approach across diverse learning contexts. Thus, our methodology stands as a promising avenue for elucidating and categorizing learning behaviors, offering implications for educational research, pedagogical practice, and personalized learning interventions.

Table 6
Viscode learning behaviors clustering.

ID	Count#	Avg. Length	Avg. Time	Avg. Error %	Comments
0	4538	583.43	1204	26.46%	Active learners with efficiency
1	3231	76.24	9733	25.92%	Inactive learners without efficiency
2	1460	171.30	3652	25.51%	Medium active learners
3	–	–	–	–	No activities during the day

Table 7
BookRoll + viscode learning behaviors clustering.

ID	Count	Avg. Length	Avg. Time	Comments
0	5974	671	13,132	High efficiency learners
1	5525	118	28,494	Medium efficiency learners
2	2567	143	37,468	Low efficiency learners
3	–	–	–	No activities during the day

Table 8
Training result with validation set scores.

Embedding	Data Source	F1	Recall	Precision	Accuracy
BERT	BR	0.56	0.55	0.59	0.60
	VS	0.93	0.92	0.95	0.93
	BR + VS	0.95	0.94	0.95	0.95

BR: BookRoll, VS: Viscode, BR + VS: Consolidation of BookRoll and Viscode.

5.3. Model 2 prediction with learning behaviors DNA

In comparing Model 1 and Model 2 for learning performance prediction, both models yield similar outcomes in terms of predictive accuracy. However, Model 2 distinguishes itself by offering a more interpretable framework. While the overall predictive performance remains consistent between the two models, Model 2 provides a deeper level of insight into the underlying factors influencing learning outcomes when we look at Learning Behaviors DNA. Its enhanced granularity and explainability afford educators and stakeholders a clearer understanding of the mechanisms driving predictions, thereby facilitating more informed decision-making processes. Consequently, despite comparable predictive efficacy, the superior explanatory power of Model 2 renders it a more valuable tool for optimizing educational interventions and fostering student success, which will be discussed in the next section.

In other similar research which leverage learning behavior to predict learning performance, Jo et al. (2014) collected and analyzed the web-log data of 43 and 29 college students from two blended learning classes, one involving online discussion-based learning and the other involving offline lecture-based learning with online supplemental tool. The study used linear multiple regression analysis and random forest analysis to develop and compare prediction models for each class. The study of (Zhang et al., 2017) presents the reference indicators of learning behavior on the online learning platform, which are derived from the whole learning process and consist of four dimensions: preparation, progress, resource learning, and test. This paper analyzes the students' behavior of viewing courseware and uses the optimized Logistic Regression to predict whether a student was excellent or not. The paper demonstrates that the selected eigenvalues can effectively predict student performance.

Another study like Lee et al. (2021) collected data from two courses conducted on the National Tsing Hua University's MOOCs platform: Introduction to IoT and Calculus I. The data included students' video playback actions, such as play, pause, search, and adjust playback speed, as well as their answers to exercises and quizzes. The study used two deep neural network models to analyze the data and predict students' scores and performance. Note that most of the previous study extract the features through system log abstraction where potentially key features may be ignored and impact the prediction accuracy. In our study, both Model 1 and Model 2 use the embedding from the whole learning description document which we believe most of the learning activity semantics will be included thus a better prediction result can be demonstrated.

5.4. Learning behaviors DNA implication and response to RQ3

When delving deeper into the behaviors of top-performing students,

a consistent trend emerges in their approach to learning. These individuals exhibit a proactive stance towards their education, characterized by regular engagement during class sessions (type = 0) and a diligent commitment to completing homework or reviewing material promptly thereafter (type = 0 or 1). Notably, some of these high achievers also allocate time for review sessions during the week or even on the day preceding the class, demonstrating their strategic planning and proactive attitude towards consolidating their understanding. This disciplined and structured approach not only reflects their dedication to academic excellence but also underscores the importance of regular reinforcement and active participation in the learning process. By consistently reinforcing their understanding of course material through a combination of in-class engagement and timely review, these students effectively optimize their learning outcomes and position themselves for success.

What constitutes the optimal learning behaviors DNA for achieving a high learning outcome score? We created a learning behaviors DNA as the sequence to be composed of all '3', means the student has no interaction (did not do anything during the class) and the model forecast result is 0.036. This result indicates that students who have no interaction with the systems are likely to receive low scores. We also tested the same learning behaviors DNA sequences for '2', '1', and '0'. The forecast result we got is 0.022, 0.028 and 0.172. While '0' represents active learning behaviors and yields the highest forecast number (0.172), it does not necessarily indicate that students will achieve high scores solely by engaging actively with the systems. This is primarily because the model learned patterns from the pedagogical scenario. Further investigation into scenarios such as students' self-learning may provide additional insights into the optimal learning behaviors DNA.

Consider a scenario where a student is active during class and engages in review activities the following day. The resulting learning behaviors DNA for the entire week would be [0, 1, 3, 3, 3, 3, 3]. We expand this behavior for the entire semester and adjust the learning behaviors for the last three days to 0, indicating that the student is actively reviewing and preparing for the final examination., which means the student actively review and prepared for the final examination. When we input this learning behaviors DNA into the model for forecasting, we obtain a forecasted value of 2.26, suggesting that this type of learning behavior is associated with a high-performance outcome.

In conclusion, our study makes a significant contribution to the understanding of learning behaviors and their impact on academic outcomes through responding to **RQ3**. By introducing the concept of learning behaviors DNA and employing transformers techniques and K-means clustering, we reveal the intricate relationship between daily behaviors and learning performance. Our findings not only affirm the importance of proactive engagement and strategic learning practices but also highlight the nuanced nature of this relationship, considering factors such as the level of interaction with educational systems. Moreover, our application-oriented approach demonstrates the practical relevance of our framework in real-world educational contexts, offering actionable insights for educators and stakeholders seeking to optimize student outcomes.

6. Conclusions

In this study, our NLP-based approach aims to revolutionize learning performance prediction by analyzing complex student interactions. We introduce Learning Behaviors DNA, a transformative framework that encapsulates student learning behaviors. We utilize HuggingFace pre-trained BERT, Google Gemini, and OpenAI to convert raw text from daily learning descriptions and semester-long interactions into rich embeddings, achieving precise performance prediction results. In the daily learning descriptions embedding, This DNA serves as a compact representation, capturing nuances like study habits, engagement trajectories, and evolving performance patterns.

Utilizing Learning Behaviors DNA, we accurately predict student

learning outcomes and uncover various learning patterns. By integrating diverse embedding models, we decode the intricate relationships between behavioral features and academic success. We adapt our embedding models accordingly by different text length. In adapting to varying text lengths, we utilize HuggingFace pre-trained BERT for shorter texts in daily behaviors clustering, while turning to OpenAI for longer documents. In assessing the cost-effectiveness of utilizing open source versus paid solutions for learning analytics, the comparison between HuggingFace's open-source transformer and OpenAI's large text embedding service reveals a significant advantage for open-source options.

While both HuggingFace and OpenAI offer state-of-the-art technologies for natural language processing tasks, the adoption of open-source solutions presents a more economically sustainable approach. By leveraging HuggingFace's open-source transformer, educational institutions can achieve comparable levels of accuracy in learning performance prediction while significantly reducing costs associated with licensing fees or usage charges. Furthermore, the collaborative nature of open-source development fosters continuous improvement and adaptation to diverse educational contexts, ensuring long-term scalability and relevance. Therefore, from a cost-effective perspective, embracing open-source solutions such as HuggingFace's transformer emerges as a prudent choice for institutions seeking to maximize the value of their learning analytics investments without compromising on predictive accuracy or functionality.

This study demonstrates the effectiveness of natural language as a superior input for performance prediction and envisioning a future where teachers may verbally describe student behaviors and our model accurately predicts performance, Learning Behaviors DNA emerges as a guiding principle, directing educators towards personalized, data-informed pedagogy.

Future work involves exploring additional models, comparing their embedding results, and refining our predictions. There are couple future research areas which can be further developed. The further exploration of the integration of NLP and LLM methodologies with learning analytics may further improve the accuracy of learning performance prediction. we will investigate the effectiveness of the proposed methodology in different educational contexts with diverse course subjects and pedagogical scenarios and examine the long-term impact on student learning and academic achievement. Furthermore, the learning behaviors DNA can also be utilized in the development of personalized interventions and support strategies based on the DNA to enhance student outcomes.

In summary, our innovative NLP-driven approach, powered by Learning Behaviors DNA, serves as a transformative bridge between raw logs and actionable insights. This methodology empowers educators, elevates student outcomes, and propels education into an era defined by precision and personalization.

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Declaration of open data, ethics and conflict of interest

The study was approved by an ethical committee with ID: 201812ES208 and 202005ES032. Informed consent was obtained from all participants, and their privacy rights were strictly observed. The participants were protected by hiding their personal information during the research process. They knew that the participation was voluntary, and they could withdraw from the study at any time. There is no potential conflict of interest in this study. The data can be obtained by sending request e-mails to the corresponding author.

Availability of data and materials

Not applicable.

CRedit authorship contribution statement

Chien-Chang Lin: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Eddie S.J. Cheng:** Software, Data curation. **Anna Y.Q. Huang:** Writing – review & editing, Validation, Formal analysis, Conceptualization. **Stephen J.H. Yang:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT to conduct proofreading to improve readability and language of single sentences as some authors are not native English speakers. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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